

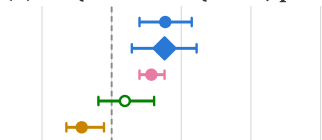
(a) Isolation gate: fresh – committed

vs commit-once · MMLB
vs commit-once · LDU
vs gold static · MMLB
vs gold static · LDU



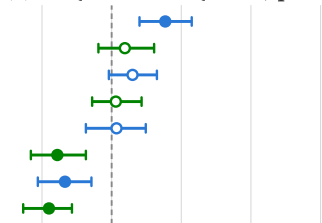
(b) PAQ-KV – ColQwen2, per dataset

MMLB (dev)
MMLB (sealed)
MMDocRAG
LDU
MMDocIR-QA



(c) PAQ-KV – ColQwen2, per reader

Qwen3-VL-8B · MMLB
Qwen3-VL-8B · LDU
Qwen3-VL-4B · MMLB
Qwen3-VL-4B · LDU
Qwen2.5-VL-7B · MMLB
Qwen2.5-VL-7B · LDU
InternVL3.5-8B · MMLB
InternVL3.5-8B · LDU



-0.1 0.0 0.1 0.2 0.3

Δ accuracy (document-clustered 95% CI)

● MMLB (MMLongBench-Doc) ● MMDocRAG
● LDU (LongDocURL) ● MMDocIR-QA
● CI excludes 0 ○ CI includes 0 ◆ sealed run